AeroqAI Whitepaper: Autonomous Agent Architecture & Governance

Section 1: Autonomous Agent Reasoning Protocol & ReAct Loop

This document specifies runtime controls and execution limits for autonomous LLM agents.

1.1 Recursion Depth Constraints: The maximum reasoning loop recursion depth is strictly capped at 8 iterations per

If an agent fails to synthesize a final conclusion within 8 iterative hops, execution halts with a MaxHopsExceeded ale

1.2 Schema Validation: All tool calls generated during the ReAct loop require JSON-RPC 2.0 schema validation befo

Malformed tool arguments trigger an automated self-healing retry prompt with deterministic feedback.

Section 2: Memory Persistence & Episodic Buffer Management

Stateful agents maintain episodic and semantic memory architectures to ensure coherence across extended workflo

2.1 Sliding Window Buffer: The episodic memory buffer stores the last 50 dialogue interactions in a sliding window w

Interactions older than 50 turns are summarized into compact semantic memory anchors.

2.2 Associative Recall: Long-term associative recall uses cosine similarity with a retrieval threshold of 0.82 across ve

Memories below 0.82 similarity are suppressed to prevent distracting the central reasoning context.

Section 3: Safety Sandbox & Tool Execution Firewalls

Agent tool execution occurs in quarantined runtime environments to mitigate security vulnerabilities.

3.1 Destructive API Guardrails: Destructive API calls (DELETE, DROP, KILL, TERMINATE) require explicit human-i

Execution of destructive operations without a valid cryptographic token results in immediate process revocation.

3.2 Subprocess Timeout: Execution timeout for isolated sandboxed Python subprocesses is set to 15,000 millisecon

Processes exceeding 15 seconds are terminated by the kernel watchdog to prevent infinite loops.